# Caloric restriction, the traditional Okinawan diet, and healthy aging: the diet of the world's longest-lived people and its potential impact on mor...

Long-term caloric restriction (CR) is a robust means of reducing age-related diseases and extending life span in multiple species, but the effects in humans are unknown. The low caloric intake, long life expectancy, and the high prevalence of centenarians in Okinawa have been used as an argument to support the CR hypothesis in humans. However, no long-term, epidemiologic analysis has been conducted on traditional dietary patterns, energy balance, and potential CR phenotypes for the specific cohort of Okinawans who are purported to have had a calorically restricted diet. Nor has this cohort's subsequent mortality experience been rigorously studied. Therefore, we investigated six decades of archived population data on the elderly cohort of Okinawans (aged 65-plus) for evidence of CR. Analyses included traditional diet composition, energy intake, energy expenditure, anthropometry, plasma DHEA, mortality from age-related diseases, and current survival patterns. Findings include low caloric intake and negative energy balance at younger ages, little weight gain with age, life-long low BMI, relatively high plasma DHEA levels at older ages, low risk for mortality from age-related diseases, and survival patterns consistent with extended mean and maximum life span. This study lends epidemiologic support for phenotypic benefits of CR in humans and is consistent with the well-known literature on animals with regard to CR phenotypes and healthy aging.